

Maximum Lifetime Routing in Wireless Sensor Networks

Jae-Hwan Chang, *Member, IEEE*, and Leandros Tassiulas, *Member, IEEE*

Abstract—A routing problem in static wireless ad hoc networks is considered as it arises in a rapidly deployed, sensor based, monitoring system known as the wireless sensor network. Information obtained by the monitoring nodes needs to be routed to a set of designated gateway nodes. In these networks, every node is capable of sensing, data processing, and communication, and operates on its limited amount of battery energy consumed mostly in transmission and reception at its radio transceiver. If we assume that the transmitter power level can be adjusted to use the minimum energy required to reach the intended next hop receiver then the energy consumption rate per unit information transmission depends on the choice of the next hop node, i.e., the routing decision. We formulate the routing problem as a linear programming problem, where the objective is to maximize the network lifetime, which is equivalent to the time until the network partition due to battery outage. Two different models are considered for the information-generation processes. One assumes constant rates and the other assumes an arbitrary process. A shortest cost path routing algorithm is proposed which uses link costs that reflect both the communication energy consumption rates and the residual energy levels at the two end nodes. The algorithm is amenable to distributed implementation. Simulation results with both information-generation process models show that the proposed algorithm can achieve network lifetime that is very close to the optimal network lifetime obtained by solving the linear programming problem.

Index Terms—Energy-sensitive routing, power aware routing, wireless ad hoc networks, wireless sensor networks.

I. INTRODUCTION

CONSIDER a wireless network of static nodes randomly distributed as depicted in Fig. 1, where each node operates on limited battery energy consumed mostly in transmission and reception of data at its radio transceiver. Assume that at each node some type of information is generated and the information needs to be delivered to a set of designated gateway nodes possibly using multiple hops. The transmitter power level

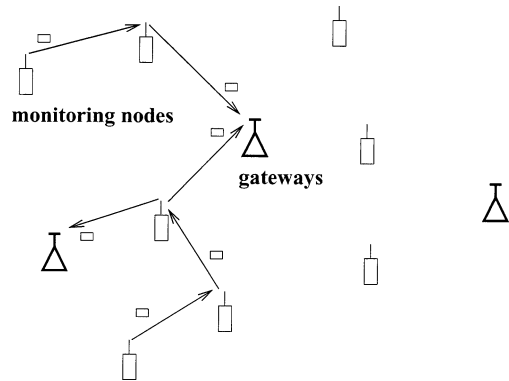


Fig. 1. A wireless sensor network is depicted where the nodes are randomly distributed and the information generated at the monitoring nodes are to be delivered to the gateway nodes.

is assumed to be adjusted to the minimum level appropriate for the intended receiver within the transmission range. Note that the routing decision and the transmission energy level selection are intrinsically connected in these power-controlled wireless ad hoc networks since the power level will be adjusted depending on the choice of the next hop node.

An example scenario for this type of wireless ad hoc network may include a wireless sensor network where the sensors gather acoustic, magnetic, or seismic information and send the information to its gateway node which has more processing power for further processing of the information or has larger transmission range for the delivery of the information to a possibly larger network for retrieval by a remote user.

Most of the previous works on routing in wireless ad hoc networks deal with the problem of finding and maintaining correct routes to the destination during mobility and changing topology [1], [7], [22]. In [1] and [7], the authors presented a simply implementable algorithm which guarantees strong connectivity and assumes limited node range. Shortest path algorithm is used in this strongly connected backbone network. In [22], the authors developed a dynamic routing algorithm for establishing and maintaining connection-oriented sessions which uses the idea of predictive re-routing to cope with the unpredictable topology changes. Some other routing algorithms in mobile wireless networks can be found in [14], [23], [25], and [26], which, as the majority of routing protocols in mobile ad hoc networks do, use shortest path or minimum hop (MH) routing.

Power consumption in the wireless ad hoc networks can be largely categorized into two parts. One is communication related and the other is noncommunication related such as processing or sensing (in case of a sensor network). A model for evaluating the communication related energy consumption

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J.-H. Chang was with the Institute for Systems Research and the Department of Electrical and Computer Engineering, University of Maryland, College Park, MD 20742 USA. He is now with Telecommunication R&D Center, Samsung Electronics Company, Ltd., Suwon 442-600, Korea (e-mail: jchang@ieee.org).

L. Tassiulas was with the Institute for Systems Research and the Department of Electrical and Computer Engineering, University of Maryland, College Park, MD 20742 USA. He is now with Computer Engineering and Telecommunications, University of Thessaly, Volos, Greece (e-mail: leandros@uth.gr)

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behavior of a mobile ad hoc network was presented in [9], where the power consumption was further categorized into three modes: transmission, reception, and idle mode. Power saving in idle mode was studied in [36], which we believe is complementary to our work. However, we, as well as the others mentioned in the following, have been focusing in power savings during transmission and reception. In [11], the sum of multiples of the transmission power and the power price was proposed as the path length to be minimized. The power price was a function of the current battery level, total storage capacity, type of battery, etc., but they did not specify the metric. In [32], the conditional max-min battery capacity routing (CMMBCR) was proposed, which is a combination of minimum total energy (MTE) routing and max-min residual energy routing. The minimum total transmission and reception energy path is chosen in the set of all paths whose minimum residual energy is above a given threshold. If the set is empty then max-min residual energy route is used. In [18], routing for the maximum network lifetime was studied, where the message sequence is not known *a priori*. An approximation algorithm called $\max - \min zP_{\min}$ was proposed which tries to strike a balance between the minimum transmission energy routing and the max-min residual energy routing. Scalability of the algorithm was provided by zone-based hierarchical routing approach. Reception energy consumption is assumed to be included in the transmission energy consumption since all intermediate nodes except the source and the destination are engaged in both transmission and reception. The performance of their algorithm was shown to be close to the optimal solution obtained by linear programming. The $\max - \min zP_{\min}$ algorithm first finds the minimum transmission energy path (let the total transmission energy on this path be $P_{\min}$) and then removes all edges whose residual energy fraction after use is smaller than or equal to the minimum residual energy fraction on the minimum transmission energy path. It then repeats the same procedure on the subgraph until just before the total transmission energy of the chosen path exceeds z times $P_{\min}$, where $z \geq 1$. The resulting path is assigned to the incoming traffic. In [31], a routing metric similar to ours have been used. However, instead of requiring the global network information they only require localized routing information and assume that nonlocal routing information can be treated equally in all paths. Refer to [20] for a good survey on the power optimization in routing protocols.

The problem of minimum energy routing has been addressed before in [1], [7], [8], [10], [21], and [28]–[30]. The approach in these papers, called the minimum total energy (MTE) routing here, was to minimize the total consumed energy to reach the destination, which minimizes the energy consumed per unit flow or packet. However, if all the traffic is routed through the minimum energy path to the destination, the nodes in that path will run out of batteries quickly rendering other nodes useless due to a network partition even if they do have available energy resource. In our work, instead of trying to minimize the total consumed energy on the path, the performance objective of maximizing the lifetime of the system [4], which is equivalent to maximizing the time to network partition [30] has been considered. In [4] we identified the maximum lifetime problem as a linear programming problem and in [5] the problem was ex-

tended to the multicommodity case. Since it is a linear programming problem, it is solvable in polynomial time. While in [5] constant information-generation rate case was considered, in [6] some arbitrary information-generation process model was studied.

In this paper, the maximum lifetime routing problem is extended to include the energy consumption at the receivers during reception. Note, however, that the energy consumption at the unintended receiver nodes that overhear the transmission is not included. This extension was applied to the algorithm as well. In the simulation, comparison is made with the optimal network lifetime obtained by solving the linear programming problem as well as with two other algorithms proposed in [18] and [32]. Note that due to the inherent nonscalability of table-driven routing approach, the proposed solution in its current form is not scalable and hence may not be suitable for direct application to large networks. Note also that the energy consumption due to routing control packets are not included in the model or the simulation since we assume a situation where the energy consumption is dominated by the data packets. However, in the simulation we show that there is a tradeoff between the routing information update rate and the performance so that the number of routing control packets can be reduced with some sacrifice in the performance.

Brown *et al.* [3] have extended the objective of power-aware routing in the multicommodity case to sequentially maximizing the lifetime of each commodity, while we only maximize the time until the first commodity network partition.

Information from other sources were utilized for the routing decision in the following works. Geographical information of the communication nodes is used in [15], [16], [31], [34], and [36]. Upper layer information is utilized in [12], [13], and [17].

Distributed topology control was studied in [27] and [33] where transmitter power levels are selected to guarantee the network connectivity while saving transmission energy, which can be complementary to our work. A good survey on the topology control, clustering, broadcasting, and multicasting can be found in [19].

In our study the nodes are not mobile and the topology of the network is static. Hence the results are applicable to networks which are either static, like the sensor networks we mentioned earlier, or whose topology changes slowly enough such that there is enough time for optimally balancing the traffic in the periods between successive topology changes.

This paper is organized as follows. In Section II, the maximum system lifetime routing problem is formulated for fixed information-generation rates as well as for some arbitrary information-generation process. In Section III, we propose the flow augmentation (FA) algorithm which iteratively augments traffic flow along the shortest cost path. The proposed link cost reflects both the residual energy at the transmitting node and the receiving node and the energy consumption in unit data transmission over the link. In Section IV, simulation on randomly generated graphs is performed to evaluate the performance of the proposed algorithm both for the fixed information-generation rates and for a certain scenario where information is generated at monitoring nodes that detect moving targets. Finally in Section V, some concluding remarks are made.

II. ROUTING FOR THE MAXIMUM SYSTEM LIFETIME

In this section, we first formulate the maximum system lifetime routing problem for the case where information-generation rates are fixed. Then, we consider a more general case where we are given some arbitrary information-generation processes instead of fixed information-generation rates.

A. Constant Information Generation Rates

A wireless sensor network is modeled as a directed graph $G(N, A)$ where N is the set of all nodes and A is the set of all directed links (i, j) where $i, j \in N$. Link (i, j) exists if and only if $j \in S_i$, where S_i is the set of all nodes that can be directly reached by node i with a certain transmit power level in its dynamic range. Each node i has the initial battery energy of E_i . The transmission energy consumed at node i to transmit a data unit to its neighboring node j is denoted by e_{ij}^t and the energy consumed by the receiver j is denoted by e_{ji}^r . Let there be multiple commodities where a commodity is defined by a set of source nodes and destination nodes. We are given, for each commodity $c \in C$, a set of origin nodes $O^{(c)}$ where information is generated at node i with rate $Q_i^{(c)}$, i.e.,

$$O^{(c)} = \{i | Q_i^{(c)} > 0, i \in N\} \quad (1)$$

and a set of destination nodes $D^{(c)}$ among which any node can be reached in order for the information transfer of commodity c to be considered done. Let $q_{ij}^{(c)}$ be the transmission rate of commodity c from node i to node j to be assigned by the routing algorithm.

The lifetime of node i under a given flow $\mathbf{q} = \{q_{ij}^{(c)}\}$ is given by

$$T_i(\mathbf{q}) = \frac{E_i}{\sum_{j \in S_i} e_{ij}^t \sum_{c \in C} q_{ij}^{(c)} + \sum_{j: i \in S_j} e_{ji}^r \sum_{c \in C} q_{ji}^{(c)}}. \quad (2)$$

Now, let us define the *system lifetime* or the *network lifetime* under flow $\mathbf{q}$ as the minimum lifetime over all nodes, i.e.,

$$T_{sys}(\mathbf{q}) = \min_{i \in N} T_i(\mathbf{q}). \quad (3)$$

Note that in our flow model with fixed information-generation rates, the system lifetime is equivalent to the earliest network partition time of a commodity and is by definition the time of the first node death.

Our goal is to find the flow that maximizes the system lifetime under the flow conservation condition. Note that maximizing the system lifetime is equivalent to maximizing the amount of total information transfer given a fixed information-generation rates. The problem can be written as follows.

$$\begin{aligned} & \text{Maximize } T_{sys}(\mathbf{q}) \\ & \text{s.t. } q_{ij}^{(c)} \geq 0, \quad \forall i \in N, \forall j \in S_i, \forall c \in C, \\ & \sum_{j: i \in S_j} q_{ji}^{(c)} + Q_i^{(c)} = \sum_{j \in S_i} q_{ij}^{(c)}, \quad \forall i \in N - D^{(c)}, \forall c \in C. \end{aligned} \quad (4)$$

Fig. 2 illustrates the flow conservation condition for commodity c at node i , and it should be noted that the condition applies to each commodity separately.

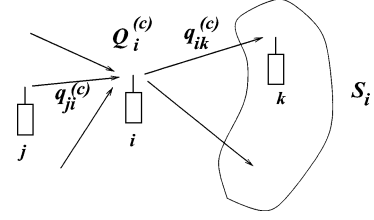


Fig. 2. Conservation of flow condition at node i for each commodity c requires that the sum of information-generation rate and the total incoming flow must equal the total outgoing flow.

In the following we show that the problem is a linear programming problem [24]. The problem of maximizing the system lifetime, given the information-generation rates $Q_i^{(c)}$ at the set of origin nodes $O^{(c)}$ and the set of destination nodes $D^{(c)}$ for each commodity c , is equivalent to the following linear programming problem:

$$\begin{aligned} & \text{Maximize } T \\ & \text{s.t. } \hat{q}_{ij}^{(c)} \geq 0, \quad \forall i \in N, \forall j \in S_i, \forall c \in C, \\ & \sum_{j \in S_i} e_{ij}^t \sum_{c \in C} \hat{q}_{ij}^{(c)} + \sum_{j: i \in S_j} e_{ji}^r \sum_{c \in C} \hat{q}_{ji}^{(c)} \leq E_i, \quad \forall i \in N, \\ & \sum_{j: i \in S_j} \hat{q}_{ji}^{(c)} + T Q_i^{(c)} = \sum_{j \in S_i} \hat{q}_{ij}^{(c)}, \quad \forall i \in N - D^{(c)}, \forall c \in C \end{aligned} \quad (5)$$

where $\hat{q}_{ij}^{(c)} = T q_{ij}^{(c)}$ is the amount of information of commodity c transmitted from node i to node j until time T . Note that the variable T in (5) should be considered as an independent variable in order to see the equation as a linear programming problem.

B. Arbitrary Information Generation Processes

In this section, a more practical scenario than the one introduced in the previous section will be considered for wireless sensor networks. Instead of having a fixed set of origin nodes with fixed information-generation rates, a packet is generated periodically¹ at each sensor node if the sensor is detecting a moving target. For simplicity, we assume that there is only one commodity in this case, which means that all information has the same set of destination nodes.

Notice the difference between the problem discussed here with the previous one. In the previous problem, we were given fixed information-generation rates, which implied that the amount of information generated in some time interval T is known *a priori*. On the contrary, here we assume that the amount of total information generated in some time interval T is not known *a priori* but we try to make routing decision on the fly as new information is generated. In this scenario, a number of sensors are randomly distributed, and target objects move about in or pass through the region. Each sensor generates a packet periodically if and only if there is any target object in its sensor range. The generated packets are to be delivered to one of the designated gateway nodes. Our goal is to select the route of each generated packet such that the time until the first failure of the packet delivery due to battery outage is maximized.

¹Although it does not have to be periodic, it is assumed so for simplicity. In case it is not periodic, the lifetime will not be readily given in absolute time units such as in seconds but in number of possibly unequal discrete time units.

In the following paragraph, we will describe an integer programming problem the solution of which will be used as a performance bound for the problem that we are trying to solve. As mentioned earlier, the problem we are interested in assumes no knowledge about the future information-generation processes. However, for the following performance bound we will assume the perfect knowledge of the future information-generation processes.

Let's consider the feasibility problem first. We would like to determine if the information generated until time T can be delivered to one of the set of destination nodes D . Let $\hat{Q}_i(T)$ be the number of packets generated at origin node $i \in O$ during the time interval $[0, T)$, and let $\hat{q}_{ij}(T)$ be the total number of packets routed through link $(i, j) \in A$. It is feasible if there exists a set of nonnegative integers $\hat{q}_{ij}(T)$ for each link $(i, j) \in A$ which satisfies the following two conditions. The conservation of flow condition is given by

$$\sum_{j \in S_j} \hat{q}_{ji}(T) + \hat{Q}_i(T) = \sum_{j \in S_i} \hat{q}_{ij}(T), \quad \forall i \in N - D \quad (6)$$

and the total energy constraint is given by

$$\sum_{j \in S_i} e_{ij}^t \hat{q}_{ij}(T) + \sum_{j \in S_i} e_{ji}^r \hat{q}_{ji}(T) \leq E_i, \quad \forall i \in N \quad (7)$$

where e_{ij}^t and e_{ji}^r are the energy consumption in transmitting and receiving one packet over the link (i, j) at nodes i and j respectively. Our goal in terms of this feasibility problem can be stated as finding the maximum feasible time T .

Note that the problem with constant information-generation rates in the previous section is a special case of this more general formulation.

In the following, we discuss conditions for the feasibility. For a set of nodes V , assume that each node $i \in V$ has the amount of information generated during $[0, T)$, $\hat{Q}_i(T)$, which needs to be delivered out of V . For a node $i \in V$ let e_i^V be the least energy expenditure for transporting an information unit out of V . If there is no outgoing link of i through which information can be forwarded out of V , $e_i^V = \infty$. Assume here that no energy is consumed in reception for the simplicity of the discussion. The necessary feasibility condition is given by

$$\sum_{i \in V} \hat{Q}_i(T) \leq \sum_{i \in V} \frac{E_i}{e_i^V} \quad (8)$$

which states that the total information generated should not be greater than the capacity of all outgoing flow paths. Note that E_i/e_i^V is the maximum amount of information that can flow out of V via node i .

The following counterexample in Fig. 3 shows that the necessary condition above is not sufficient. One can verify that the necessary feasibility condition is met. However, the flow is not feasible since the total energy constraint at node a corresponding to (7)

$$0.5\hat{q}_{ab}(T) + (4 - \hat{q}_{ab}(T)) \leq 2.5 \quad (9)$$

requires $\hat{q}_{ab}(T) \geq 3$, and the total energy constraint at node b requires $\hat{q}_{bd}(T) \leq 2$, but at the same time $\hat{q}_{ab}(T) = \hat{q}_{bd}(T)$

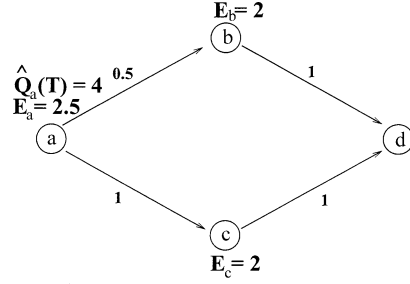


Fig. 3. Counterexample showing that the necessary feasibility condition is not sufficient. The numbers next to the links are the energy expenditure per data unit transmitted across the link.

should hold according to the flow conservation condition, which is impossible.

It can be verified that if the energy expenditure through all the outgoing links of a node were the same then the necessary condition would be sufficient as well. In other words, if the transmit power levels are fixed, then the condition becomes both necessary and sufficient for feasibility.

III. FLOW AUGMENTATION ALGORITHM

In this section, we propose a heuristic called the flow augmentation (FA) algorithm which is an extension to what has been presented in [5]. We will describe the algorithm for fixed information-generation rates.

A high level description of the algorithm is given in the following for fixed information-generation rates. At each iteration, each origin node $o \in O^{(c)}$ of commodity c calculates the shortest cost path to its destination nodes in $D^{(c)}$, where the cost will be defined later. Then the flow is augmented by an amount of $\lambda Q_i^{(c)}$ on the shortest cost path, where λ is the augmentation step size which is equivalent to the amount of information routed between routing information updates. For example, if the routing information is updated after every packet is routed then this value represents the packet size. Residual energy at each node is updated just before each routing information update, which will change link costs. With the updated link costs, the shortest cost paths are recalculated and the procedures are repeated until any node $i \in N$ runs out of its initial total energy E_i .

Our objective is to find the best link cost function which will lead to the maximization of the system lifetime. There are some parameters to consider in calculating the link cost $cost_{ij}$ for link (i, j) . They are the energy expenditure for unit data transmission over the link, e_{ij}^t and e_{ji}^r , the initial energy E_i and E_j , and the residual energy, $\underline{E}_i$ and $\underline{E}_j$. A good candidate for the flow augmenting path should consume less energy and should avoid nodes with small residual energy since we would like to maximize the minimum lifetime over all nodes. In [30], each of these parameters were separately considered, but the combinations of them were not. We propose a new link metric which combines these parameters in one. In the beginning when all the nodes have plenty of energy, the minimum total consumed energy path is desired, while as residual energy decreases it is more important to avoid the nodes with small residual energy. Therefore, the link cost function should be such that when the nodes have

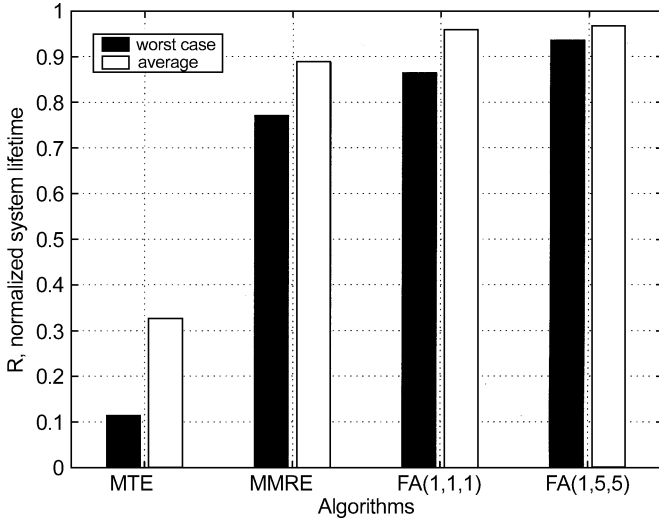


Fig. 18. Performance comparison of the algorithms in some arbitrary information-generation scenario is shown.

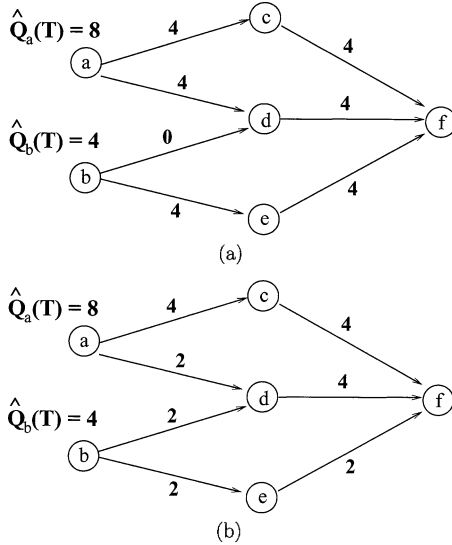


Fig. 19. An example network showing that the performance of the algorithm depends on the information-generation sample paths. Numbers on the links indicate the number of packets routed through the links. (a) When eight packets at node a are generated before the four packets at node b , the algorithm achieves the optimal system lifetime of 12 time units as shown. (b) When four packets at node b are generated before the eight packets at node a , the system lifetime is 10 time units leaving two undeliverable packets at node a .

From the above simulation results, we found out that for some information-generation scenarios it is possible to make routing decision on-the-fly and obtain close-to-optimal system lifetime. However, this may not always be the case. Actually, considering the fact that we assume no *a priori* knowledge about the future information-generation process, the simulation results are too good to believe. In the following, we give an example that shows that the performance of the algorithm depends on the information-generation sample paths of the process. Consider a network in Fig. 19 where each node has four units of energy. It requires one unit of energy per packet to cross each link except for two links, $e_{ac}^t = e_{ad}^t = 0.5$. The reception energy consumption is assumed to be zero. If eight packets are generated at node a before the four packets are generated at node b , the algorithm

finds the routes as shown in Fig. 19(a) which achieves the optimal system lifetime of 12 time units. However, if four packets at node b are generated before the eight packets at node a , the algorithm will split the traffic generated at node b equally to node d and node e and hence use the energy at node d which should have been dedicated solely to the information generated at node a in order to achieve the optimal system lifetime.

V. CONCLUSION

In wireless sensor networks where nodes operate on limited battery energy, the efficient utilization of the energy is very important. One of the main characteristics of these networks is that the transmission power consumption is closely coupled with the route selection. The energy efficiency has been considered in wireless ad hoc network routing, but the conventional routing objective was to minimize the total consumed energy in reaching the destination. In this paper, we have formulated the routing problem as maximizing the network lifetime. The new problem formulation has revealed that the minimum total energy (MTE) routing is not suitable for network-wise optimum utilization of transmission energy. We showed that significant improvement can be made by the newly proposed routing algorithm in terms of maximizing the system lifetime, which can also be interpreted as maximizing the amount of information transfer between the origin and destination nodes given the limited energy. The proposed algorithm is a shortest cost path routing whose link cost is a combination of transmission and reception energy consumption and the residual energy levels at the two end nodes. The simulation results showed close-to-optimal performance most of the time with both the fixed information-generation rates and some arbitrary information-generation process of a moving target detecting scenario in wireless sensor networks. Future research directions will be to study the effect of network density and quantized residual energy levels on the performance and overhead of the algorithm, to apply the new link metric on the on-demand routing protocols, and to consider medium access layer issues.

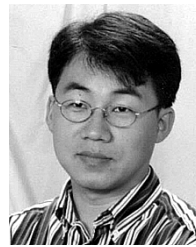
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Jae-Hwan Chang (S'92–M'03) was born in Busan, Korea, in 1970. He received the B.S. and M.S. degrees, both in electronics engineering, from Seoul National University, Seoul, Korea, in 1992 and 1994, respectively, and the Ph.D. degree in electrical and computer engineering from the University of Maryland, College Park, in 2003.

From 1994 to 1995, he was an Engineer for the R&D Center, Modacom Company, Ltd., Seoul. In 2000, he worked as an intern in Telcordia's Applied Research, Morristown, NJ. In 2003, he joined the Telecommunication R&D Center, Samsung Electronics Company, Ltd., Suwon, Korea, where he is currently a Senior Engineer. His research interests include power control, smart antennas, adaptive modulation and coding, and wireless ad hoc networking.



Leandros Tassiulas (S'89–M'91) was born in 1965 in Katerini, Greece. He received the Diploma in electrical engineering from the Aristotelian University of Thessaloniki, Thessaloniki, Greece, in 1987, and the M.S. and Ph.D. degrees in electrical engineering from the University of Maryland, College Park, in 1989 and 1991, respectively.

He is a Professor in the Department of Computer and Telecommunications Engineering, University of Thessaly, Greece, and Research Professor in the Department of Electrical and Computer Engineering and the Institute for Systems Research, University of Maryland, College Park, since 2001. He has held positions as Assistant Professor at Polytechnic University New York (1991–95), Assistant and Associate Professor, University of Maryland, College Park (1995–2001), and Professor, University of Ioannina, Greece (1999–2001). His research interests are in the field of computer and communication networks with emphasis on fundamental mathematical models, architectures and protocols of wireless systems, sensor networks, high-speed Internet and satellite communications.

Dr. Tassiulas received a National Science Foundation (NSF) Research Initiation Award in 1992, an NSF CAREER Award in 1995, an Office of Naval Research Young Investigator Award in 1997, a Bodosaki Foundation Award in 1999, and the IEEE INFOCOM'94 Best Paper Award.